

AD 2 AERODROMES

Note: The following sections in this chapter are intentionally left blank: AD-2.7, AD-2.16, AD-2.21, AD-2.22, AD-2.23, AD-2.25

RPLI AD 2.1 AERODROME LOCATION INDICATOR AND NAME

RPLI - LAOAG INTERNATIONAL AIRPORT

RPLI AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	ARP coordinates and site at AD	181034N 1203152E.
2	Direction and distance from (city)	72° 48' and 6.77 KM southwest from Laoag City proper.
3	Elevation/Reference temperature	24 FT.
4	Geoid undulation at AD ELEV PSN	119 FT.
5	MAG VAR/Annual Change	3°W (2021) / 6.0' increasing.
6	AD Operator, address, telephone, telefax, telex, AFS	Civil Aviation Authority of the Philippines Laoag International Airport Laoag City, Ilocos Norte 2900 Phone: (+63) (077) 670-8446 Email: laoag_airport@caap.gov.ph ats_area1@caap.gov.ph
7	Types of traffic permitted (IFR/VFR)	IFR-VFR.
8	Remarks	Nil.

RPLI AD 2.3 OPERATIONAL HOURS

1	AD Operator	MON - FRI: 0000 - 0900. SAT, SUN + HOL: On request.
2	Customs and immigration	2200 - 1000 (On request).
3	Health and sanitation	2200 - 1000 (On request).
4	AIS Briefing Office	Nil.
5	ATS Reporting Office (ARO)	2200 - 1400.
6	MET Briefing Office	H24.
7	ATS	2200 - 1400.
8	Fuelling	On request.
9	Handling	2200 - 1000.
10	Security	H24.
11	De-icing	Nil.
12	Remarks	Airport Operations: 2200 - 1400.

RPLI AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	Trucks and buses by arrangement.
2	Fuel/oil types	Jet A-1, AVGAS.
3	Fuelling facilities/capacity	Tank truck, drums. Service requires prior notice.
4	De-icing facilities	Nil.
5	Hangar space for visiting aircraft	KGKIRK Aeronautics.
6	Repair facilities for visiting aircraft	Nil.
7	Remarks	Ground Handling Facilities: MacroAsia Airport Services Corporation and KGKIRK Aeronautics.

RPLI AD 2.5 PASSENGER FACILITIES

1	Hotels	Available in the vicinity and city.
2	Restaurants	Available in the city.
3	Transportation	Public jeepneys available from 2200 - 1000. Cars/Buses/Taxi/Tricycle for hire.
4	Medical facilities	Hospitals with ambulance service in the city.
5	Bank and Post Office	In the city.
6	Tourist Office	In the city.
7	Remarks	Nil.

RPLI AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	AD category for fire fighting	Category 7.
2	Rescue equipment	Two (2) fire trucks [One (1) Oshkosh (6000 liters) and one (1) Oshkosh (12000 liters)].
3	Capability for removal of disabled aircraft	Aircraft recovery equipment for lifting and moving all aircraft size up to and including A320 by third party provider. Disabled Aircraft Removal Plan (DARP) Coordinators: 1. Laoag Fire Chief Phone: (+63) (077) 670-8446, +639936753551 (mobile) 2. Laoag Safety Officer Phone: (+63) (077) 670-8446, +639171025062 (mobile) 3. Laoag Airport Manager Phone: (+63) (077) 670-8446, +639688807284 (mobile) 4. DARP Third Party Contractor AB Taylan Phone: +639399165737 (mobile) KGKIRK Aeronautics Phone: +639478811180 (mobile)
4	Remarks	Foam Meeting Performance: Level B.

RPLI AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Apron surface and strength	Main Apron Surface: CONC. Strength: 608/R/B/W/U.
2	Taxiway width, surface and strength	TWY N Width: 40 M. Surface: CONC. Strength: 608/R/B/W/U. TWY S Width: 32 M. Surface: CONC. Strength: 608/R/B/W/U.
3	Altimeter checkpoint location and elevation	Nil.
4	VOR checkpoints	Nil.
5	INS checkpoints	Nil.
6	Remarks	APN Dimension: 577 M x 54 M.

RPLI AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand ID signs, TWY guide lines and visual docking/parking guidance system of aircraft stands	Taxiing guidance signs at all intersections of TWY/RWY. Guide lines at APN.
2	RWY and TWY markings and LGT	RWY: Designation, THR, center line, edge, RWY end as appropriate, marked and lighted. TWY: Center line.
3	Stop bars and RWY guard lights	Where appropriate.
4	Other RWY protection measures	Nil.
5	Remarks	Nil.

RPLI AD 2.10 AERODROME OBSTACLES

In approach/TKOF areas			In circling area and at AD		Remarks
1			2		3
RWY/Area affected	Obstacle type Elevation Markings/LGT	Coordinates	Obstacle type Elevation Markings/LGT	Coordinates	
a	b	c	a	b	
01/APCH	Electrical Post 55 FT	180941.2N 1203149.0E	Terrain 3754 FT	173852.4N 1203027.5E	Nil
	Electrical Post 54 FT	180939.8N 1203148.8E	Terrain 3754 FT	173856.8N 1203031.9E	
	Electrical Post 52 FT	180943.1N 1203149.3E	Terrain 2799 FT	174734.3N 1203157.5E	
			Terrain 670 FT	182731.4N 1203529.3E	
			Radio Antenna 412 FT	181229.7N 1203233.2E	
			Antenna Tower 391 FT	181240.7N 1203231.0E	
			Antenna 349 FT	181226.7N 1203228.5E	
			Terrain 266 FT	175513.4N 1202450.0E	
			Tower 143 FT	180939.7N 1203138.9E	

RPLI AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Associated MET Office	Laoag Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA).
2	Hours of service MET Office outside hours	H24. -
3	Office responsible for TAF preparation Periods of validity	Laoag PAGASA. -
4	Trend forecast Interval of issuance	- -
5	Briefing/consultation provided	Personal consultation.
6	Flight documentation Language(s) used	Charts, abbreviated plain language text. English.
7	Charts and other information available for briefing or consultation	Nil.
8	Supplementary equipment available for providing information	Nil.
9	ATS units provided with information	Laoag Tower/Approach.
10	Additional information (limitation of service, etc.)	Nil.

RPLI AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations RWY NR	TRUE BRG	Dimensions of RWY (M)	Strength (PCR) and surface of RWY and SWY	THR coordinates RWY end coordinates THR geoid undulation	THR elevation and highest elevation of TDZ of precision APP RWY	Slope of RWY-SWY
1	2	3	4	5	6	7
01	005.40°	2784 X 45	608/R/B/W/U CONC+ASPH	180948.98N 1203147.47E 119 FT	THR 23 FT TDZ 23 FT	0.035% downhill towards THR 19
19	185.40°	2784 X 45	608/R/B/W/U CONC+ASPH	181119.14N 1203156.40E 119 FT	THR 21 FT TDZ 21 FT	
SWY dimensions (M)	CWY dimensions (M)	Strip dimensions (M)	RESA dimensions (M)	Location/ description of arresting system	OFZ	Remarks
8	9	10	11	12	13	14
Nil	200 X 86	2874 X 86	Nil	Nil	Nil	Nil
Nil	150 X 86	2874 X 86	Nil	Nil	Nil	Nil

RPLI AD 2.13 DECLARED DISTANCES

RWY Designator	TORA (M)	TODA (M)	ASDA (M)	LDA (M)	Remarks
1	2	3	4	5	6
01	2784	2984	2784	2784	Nil
19	2784	2934	2784	2784	Nil

RPLI AD 2.14 APPROACH AND RUNWAY LIGHTING

RWY Designator	APCH LGT type, LEN, INTST	THR LGT colour, WBAR	VASIS, (MEHT), PAPI	TDZ, LGT LEN
1	2	3	4	5
01	SALS 420 M, LIH	Green, 2	PAPI 3.0°, Left/Right, (48.72 FT)	Nil
19	MSALS 300 M, LIH	Green, 2	PAPI 3.0°, Left/Right, (47.15 FT)	Nil
RWY Centre Line LGT LEN, spacing, colour, INTST	RWY Edge LGT LEN, spacing, colour, INTST	RWY End LGT colour, WBAR	SWY LGT LEN, colour	Remarks
6	7	8	9	10
Nil	2664 M, 60 M, White/Yellow, LIH	Red, 2	Nil	SALS Six (6) sides barrettes spaced at 60 M interval from THR. 37 M long cross bar at the end of 300 M CONC RWY extension. THR Light Intensity: LIH. Six (6) THR light fixtures each wingbar. PAPI Six (6) NM. Path width: 0.5°. Vertical Obstruction Clearance: 1.2°. Horizontal Obstruction Clearance: 10° from RWY center line. RWY End Intensity: LIH. Six (6) RWY end light fixtures.
Nil	2664 M, 60M, White, LIH	Red, 2	Nil	MSALS Three (3) center line barrettes spaced at 30 M interval from THR. THR Light Six (6) THR light fixtures each wingbar. PAPI Five (5) NM, ±10° of RWY center line. RWY End Six (6) RWY end light fixtures.

RPLI AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	ABN/IBN location, characteristics and hours of operation	Location: 181046.62N 1203200.99E. Characteristics: Alternating white green - 25 flashes per minute. HR of OPS: 1000 - 2300.
2	LDI location and LGT Anemometer location and LGT	Anemometer 1: 181047.17N 1203156.79E. Anemometer 2: 181058.88N 1203205.30E.
3	TWY edge and centre line lighting	Edge: All TWY. Center line: Nil.
4	Secondary power supply/switch-over time	Secondary power supply to all lighting at aerodrome. Switch-over time of 15 seconds.
5	Remarks	Nil.

RPLI AD 2.17 ATS AIRSPACE

1	Designation and lateral limits	LAOAG AERODROME TRAFFIC ZONE (ATZ): A circle radius 5 NM centered on 181034N 1203152E (ARP). LAOAG CONTROL ZONE (CTR): A circle radius 10 NM centered on 181043.7N 1203144.6E (LAO DVOR/DME). LAOAG TERMINAL CONTROL AREA (TMA): From 190408N 1204627E - 183816N 1204431E thence along an arc 30 NM radius centered on 181043.7N 1203144.6E (LAO DVOR/DME) clockwise - 174431N 1204716E - 171900N 1205155E thence along an arc 55 NM radius centered on 181043.7N 1203144.6E (LAO DVOR/DME) clockwise - 190408N 1204627E.
2	Vertical limits	ATZ: SFC up to but excluding 2000 FT. CTR: SFC up to 1500 FT. TMA: 1500 FT to FL200 (excluding ATZ and ATS routes at FL160 and above).
3	Airspace classification	ATZ - B; CTR - D; TMA - D; ATS routes inside TMA below FL160 - D; ATS routes inside TMA above FL160 - A.
4	ATS unit call sign Language(s)	ATZ - Laoag Tower. CTR/TMA - Laoag Approach. English.
5	Transition altitude	11000 FT.
6	Hours of applicability	H24.
7	Remarks	Nil.

RPLI AD 2.18 ATS COMMUNICATION FACILITIES

Service Designation	Call Sign	Frequency	SATVOICE number	Logon address	Hours of operation	Remarks
1	2	3	4	5	6	7
TWR	Laoag Tower	118.1 MHZ 5062.5 KHZ 3608 KHZ	Nil	Nil	2200 - 1400	PRI FREQ. P/P PRI FREQ. P/P SRY FREQ.
APP	Laoag Approach	122.3 MHZ				PRI FREQ.

RPLI AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aid, MAG VAR, Type of supported OP(for VOR/ILS/MLS, give declination)	ID	Frequency	Hours of operation	Position of transmitting antenna coordinates	Elevation of DME transmitting antenna	Remarks
1	2	3	4	5	6	7
DVOR/DME	LAO	112.9 MHZ/ CH76X	H24	181043.7N 1203144.6E	100 FT	Coverage: DVOR/DME - 90 NM. 1117 M from start of RWY 19 and 242.05 M right of RWY 19 center line.

RPLI AD 2.20 LOCAL AERODROME REGULATIONS

1. Airport regulations

1.1 Avoid tight turning on the runway. Use the turn-around pad at the end of either runway.

2. Parking Bays

2.1 Assignment of Parking Bays

Bay Number	Coordinates	Aircraft Type	Restrictions/Remarks
1	Nil	A320 and lower category aircraft	Nil.
2	Nil	A320 and lower category aircraft	Nil.
3	Nil	A320 and lower category aircraft	Can accommodate A321 and B747-400 aircraft provided that bay number 4 is vacant.
4	Nil	A320 and lower category aircraft	Can accommodate A321 and B747-400 aircraft provided that bay number 3 is vacant.
5	Nil	A320 and lower category aircraft	Nil.

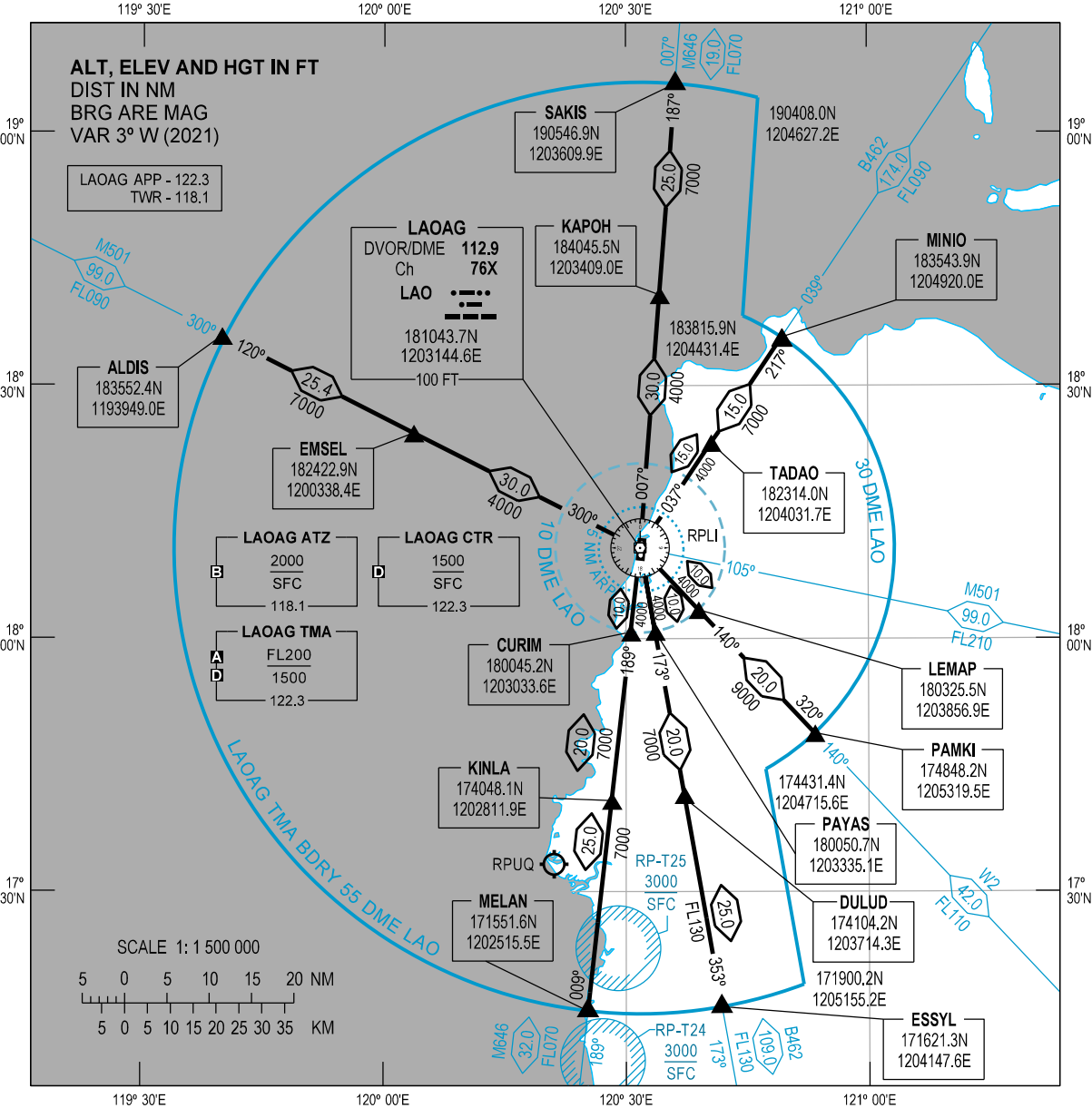
RPLI AD 2.24 CHARTS RELATED TO AN AERODROME

TITLE	PAGE
Area Chart	RPLI AD CHART 2 - 1
Standard Departure Chart - Instrument - ICAO	RPLI AD CHART 2 - 3
Standard Departure Chart - Instrument - ICAO (RNP SID RWY 01)	RPLI AD CHART 2 - 5
Standard Departure Chart - Instrument - ICAO (RNP SID RWY 19)	RPLI AD CHART 2 - 9
Standard Arrival Chart - Instrument - ICAO	RPLI AD CHART 2 - 13
Standard Arrival Chart - Instrument - ICAO (RNP STAR RWY 01)	RPLI AD CHART 2 - 15
Standard Arrival Chart - Instrument - ICAO (RNP STAR RWY 19)	RPLI AD CHART 2 - 19
Instrument Approach Chart - ICAO (VOR RWY 01)	RPLI AD CHART 2 - 23
Instrument Approach Chart - ICAO (VOR RWY 19)	RPLI AD CHART 2 - 24
Instrument Approach Chart - ICAO (RNP RWY 01)	RPLI AD CHART 2 - 25
Instrument Approach Chart - ICAO (RNP RWY 19)	RPLI AD CHART 2 - 27
Traffic Circuit Chart (RWY 01)	RPLI AD CHART 2 - 29
Traffic Circuit Chart (RWY 19)	RPLI AD CHART 2 - 30

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AREA CHART

LAOAG TMA



LEGEND		
CTA	(TMA)	
	(AWY)	
ATZ		
CRP		▲
DEPARTURE/ARRIVAL ROUTING		
DIST		25.0
MNM FLT ALT		7000
QDR		300°
COM FAILURE		
- SET TRANSPONDER MODE A CODE 7600		
- FOLLOW COM FAILURE PROC ON RELEVANT SID/STAR		

CHANGE: PAMKI COORD.

RDO NAVAID	
NAME	LAOAG
NAVAID TYPE AND FREQ	DVOR/DME 112.9 Ch 76X
IDENT	LAO ■ ■ ■ ■ ■
COORD	181043.7N 1203144.6E
ELEV OF DME ANTENNA	100 FT
NOTE:	
FOR DETAILS OF SID AND STAR PROC, SEE APPROPRIATE AD PAGES.	

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LAOAG/Laoag Intl (RPLI)
RWY 01/19
ALDIS4A/4B ESSYL4A/4B
MELAN4A/4B MINIO4A/4B
PAMKI4A/4B SAKIS4A/4B



DEPARTURE ROUTE DESCRIPTION

SID	TAKE-OFF	PROCEDURE	REMARKS
ALDIS4A	RWY 01 - Straight-out departure to 1000 FT, then left climbing turn heading 270°M.	Intercept and track-out on LAO DVOR/DME Radial 300 to EMSEL at or above 7000 FT, to ALDIS at or above 9000 FT.	PDG of 4.9% until 500 FT, then 3.3%.
ALDIS4B	RWY 19 - Straight-out departure to 1000 FT, then right climbing turn heading 332°M.		
ESSYL4A	RWY 01 - Straight-out departure to 1000 FT, then left climbing turn heading 143°M.	Intercept and track-out on LAO DVOR/DME Radial 173 to PAYAS at or above 7000 FT, to DULUD at or above FL130, to ESSYL.	Max 250 KT during initial turn. PDG of 4.9% until 500 FT, then 3.3%.
ESSYL4B	RWY 19 - Straight-out departure to 1000 FT, then right climbing turn to LAO DVOR/DME.		
MELAN4A	RWY 01 - Straight-out departure to 1000 FT, then left climbing turn heading 159°M.	Intercept and track-out on LAO DVOR/DME Radial 189 to CURIM at or above 7000 FT, to KINLA, to MELAN.	Max 250 KT during initial turn. PDG of 4.9% until 500 FT, then 3.3%.
MELAN4B	RWY 19 - Straight-out departure to 1000 FT, then right climbing turn.		
MINIO4A	RWY 01 - Straight-out departure to 1000 FT, then right climbing turn heading 067°M.	Intercept and track-out on LAO DVOR/DME Radial 037 to TADAO at or above 7000 FT, to MINIO at or above 9000 FT.	PDG of 4.9% until 500 FT, then 3.3%.
MINIO4B	RWY 19 - Straight-out departure to 1000 FT, then right climbing turn heading 067°M.		
PAMKI4A	RWY 01 - Straight-out departure to 1000 FT, then left climbing turn to LAO DVOR/DME.	Cross LAO DVOR/DME at or above 4000 FT. Track-out LAO DVOR/DME Radial 140 to LEMAP at or above 9000 FT, to PAMKI at or above 11000 FT.	Max 250 KT during initial turn. PDG of 4.9% until 500 FT, then 3.3%.
PAMKI4B	RWY 19 - Straight-out departure to 1000 FT, then right climbing turn to LAO DVOR/DME.		
SAKIS4A	RWY 01 - Straight-out departure to 1000 FT, then left climbing turn.	Intercept and track-out on LAO DVOR/DME Radial 007 to KAPOH at or above 7000 FT, to SAKIS.	PDG of 4.9% until 500 FT, then 3.3%.
SAKIS4B	RWY 19 - Straight-out departure to 1000 FT, then right climbing turn heading 037°M.		

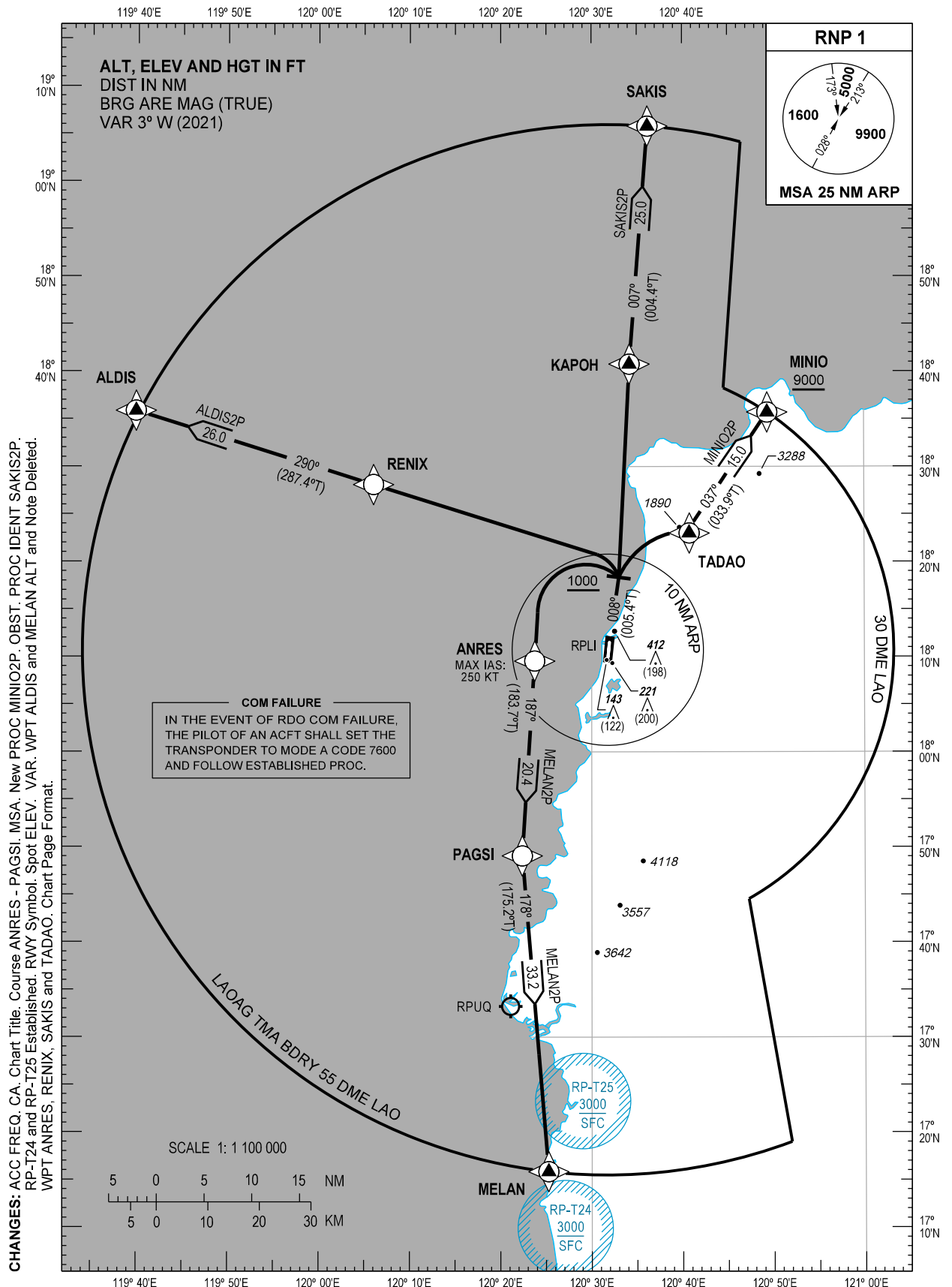
CHANGES: New PROC. Page Format.

STANDARD DEPARTURE
CHART - INSTRUMENT
(SID) - ICAO

TRANSITION
ALTITUDE
11000 FT

TWR - 118.1
APP - 122.3
ACC - 119.3 / 128.7 (NORTHWEST)
- 120.5 / 119.3 (CENTRAL NORTH)
- 8942 / 5655 (SEA-2 HF)

LAOAG/Laoag Intl (RPLI)
RNP SID RWY 01
ALDIS2P MELAN2P
MINIO2P SAKIS2P



DEPARTURE ROUTE DESCRIPTION

SID	TAKE-OFF	PROCEDURE	REMARKS
ALDIS2P	RWY 01	Climb on course 008°M. At 1000 FT, turn left direct to RENIX, to ALDIS.	PDG of 4.9% until 500 FT, then 3.3%.
MELAN2P		Climb on course 008°M. At 1000 FT, turn left direct to ANRES, to PAGSI, to MELAN.	PDG of 4.9% until 500 FT, then 3.3%. Do not exceed 250 KT until ANRES.
MINIO2P		Climb on course 008°M. At 1000 FT, turn right direct to TADAO, to MINIO at or above 9000 FT.	PDG of 4.9% until 500 FT, then 3.3%. ATS climb gradient of 4.8% to reach MINIO at or above 9000 FT.
SAKIS2P		Climb on course 008°M. At 1000 FT, turn left direct to KAPOH, to SAKIS.	PDG of 4.9% until 500 FT, then 3.3%.

WAYPOINT LIST

Waypoint Name	Latitude	Longitude	Type
ALDIS	183552.4N	1193949.0E	Fly-by
ANRES	180932.7N	1202344.0E	Fly-by
KAPOH	184045.5N	1203409.0E	Fly-by
MELAN	171551.6N	1202515.5E	Fly-by
MINIO	183543.9N	1204920.0E	Fly-by
PAGSI	174903.6N	1202222.1E	Fly-by
RENIX	182806.4N	1200559.3E	Fly-by
SAKIS	190546.9N	1203609.9E	Fly-by
TADAO	182314.0N	1204031.7E	Fly-by

PROPOSED CODING FOR DEPARTURE RNP SID RWY 01

Table 1: ALDIS2P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	008 (005.4)	+3	-	-	+1000	-	-	RNP1	RW01
20	DF	RENIX	-	-	-	-	L	-	-	-	RNP1	RW01
30	TF	ALDIS	-	290 (287.4)	+3	26.0	-	-	-	-	RNP1	RW01

Table 2: MELAN2P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	008 (005.4)	+3	-	-	+1000	-	-	RNP1	RW01
20	DF	ANRES	-	-	-	-	L	-	-250	-	RNP1	RW01
30	TF	PAGSI	-	187 (183.7)	+3	20.4	-	-	-	-	RNP1	RW01
40	TF	MELAN	-	178 (175.2)	+3	33.2	-	-	-	-	RNP1	RW01

CHANGES: New PROC MINIO2P. RTE and PROC Description. WPT List. Proposed Coding. Page Format.

Table 3: MINIO2P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	008 (005.4)	+3	-	-	+1000	-	-	RNP1	RW01
20	DF	TADAO	-	-	-	-	R	-	-	-	RNP1	RW01
30	TF	MINIO	-	037 (033.9)	+3	15.0	-	+9000	-	-	RNP1	RW01

Table 4: SAKIS2P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	008 (005.4)	+3	-	-	+1000	-	-	RNP1	RW01
20	DF	KAPOH	-	-	-	-	L	-	-	-	RNP1	RW01
30	TF	SAKIS	-	007 (004.4)	+3	25.0	-	-	-	-	RNP1	RW01

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STANDARD DEPARTURE
CHART - INSTRUMENT
(SID) - ICAO

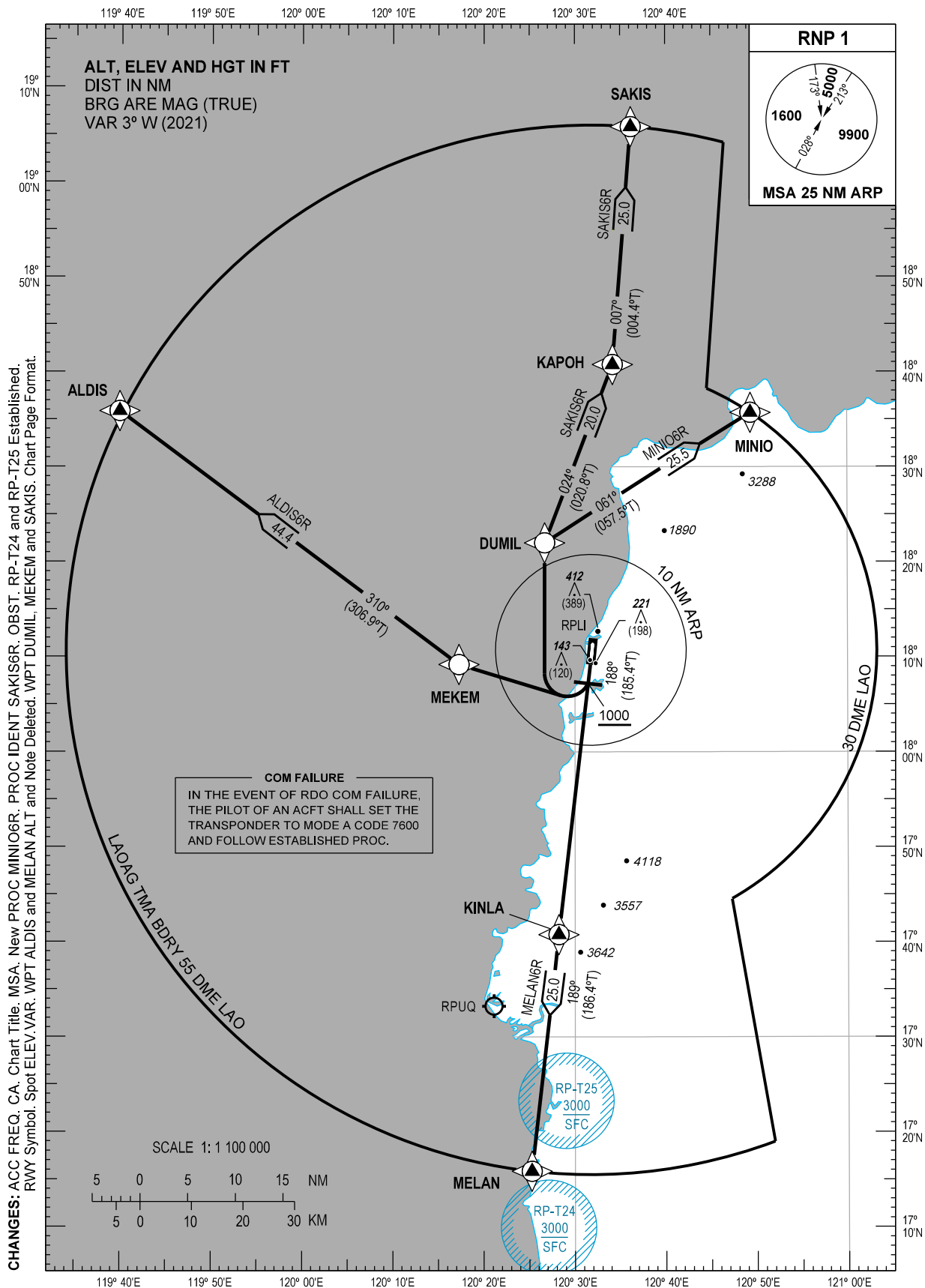
TRANSITION
ALTITUDE
11000 FT

TWR - 118.1
APP - 122.3
ACC - 119.3 / 128.7 (NORTHWEST)
- 120.5 / 119.3 (CENTRAL NORTH)
- 8942 / 5655 (SEA-2 HF)

LAOAG/Laoag Intl (RPLI)

RNP SID RWY 19

ALDIS6R MELAN6R
MINIO6R SAKIS6R



DEPARTURE ROUTE DESCRIPTION

SID	TAKE-OFF	PROCEDURE	REMARKS
ALDIS6R	RWY 19	Climb on course 188°M. At 1000 FT, turn right direct to MEKEM, to ALDIS.	
MELAN6R		Climb on course 188°M. At 1000 FT, turn right direct to KINLA, to MELAN.	
MINIO6R		Climb on course 188°M. At 1000 FT, turn right direct to DUMIL, to MINIO.	
SAKIS6R		Climb on course 188°M. At 1000 FT, turn right direct to DUMIL, to KAPOH, to SAKIS.	

WAYPOINT LIST

Waypoint Name	Latitude	Longitude	Type
ALDIS	183552.4N	1193949.0E	Fly-by
DUMIL	182200.0N	1202639.8E	Fly-by
KAPOH	184045.5N	1203409.0E	Fly-by
KINLA	174048.1N	1202811.9E	Fly-by
MEKEM	180911.7N	1201713.7E	Fly-by
MELAN	171551.6N	1202515.5E	Fly-by
MINIO	183543.9N	1204920.0E	Fly-by
SAKIS	190546.9N	1203609.9E	Fly-by

PROPOSED CODING FOR DEPARTURE RNP SID RWY 19

Table 1: ALDIS6R

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	188 (185.4)	+3	-	-	+1000	-	-	RNP1	RW19
20	DF	MEKEM	-	-	-	-	R	-	-	-	RNP1	RW19
30	TF	ALDIS	-	310 (306.9)	+3	44.4	-	-	-	-	RNP1	RW19

Table 2: MELAN6R

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	188 (185.4)	+3	-	-	+1000	-	-	RNP1	RW19
20	DF	KINLA	-	-	-	-	R	-	-	-	RNP1	RW19
30	TF	MELAN	-	189 (186.4)	+3	25.0	-	-	-	-	RNP1	RW19

Table 3: MINIO6R

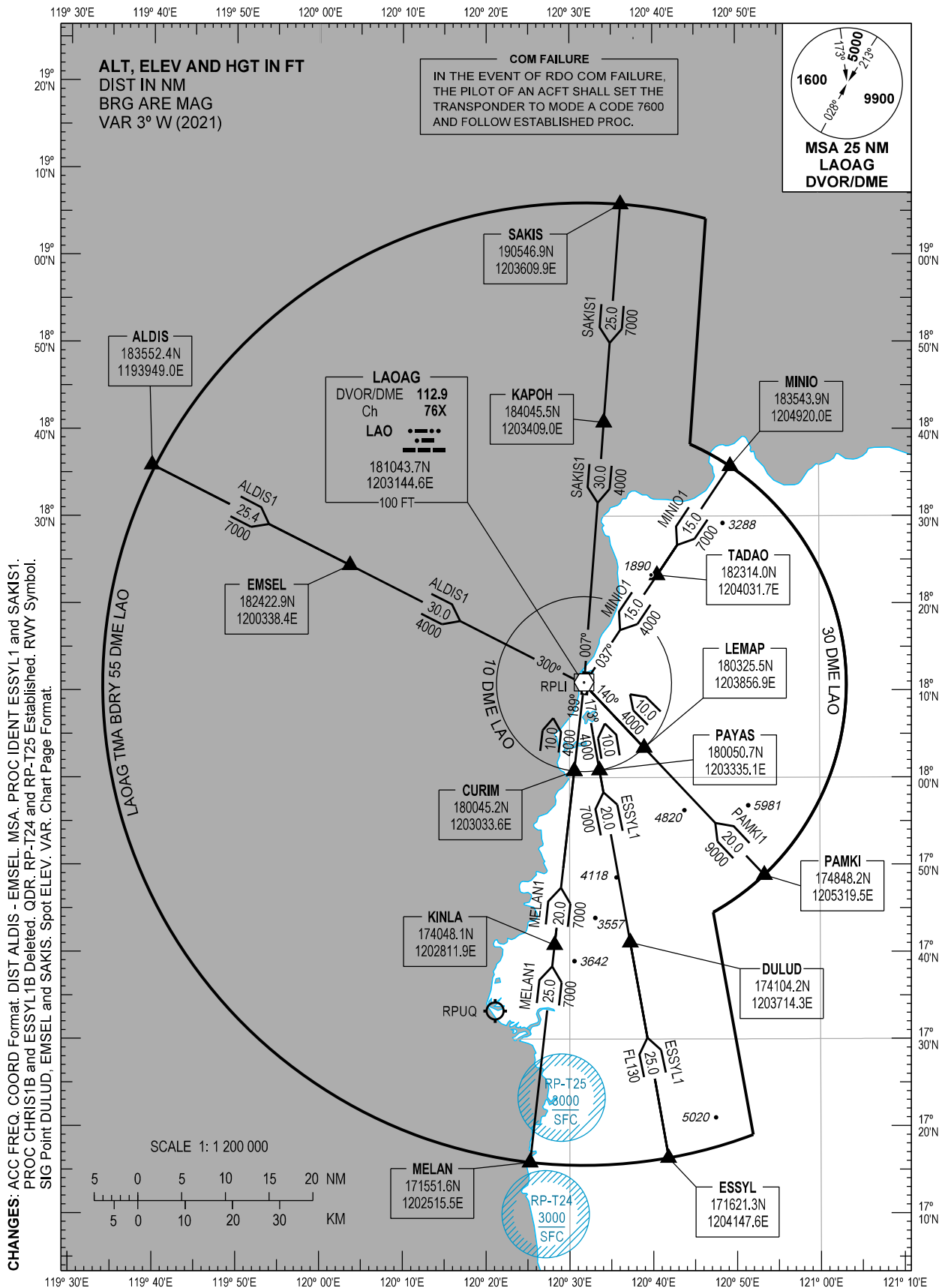
Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	188 (185.4)	+3	-	-	+1000	-	-	RNP1	RW19
20	DF	DUMIL	-	-	-	-	R	-	-	-	RNP1	RW19
30	TF	MINIO	-	061 (057.5)	+3	25.5	-	-	-	-	RNP1	RW19

Table 4: SAKIS6R

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	CA	-	-	188 (185.4)	+3	-	-	+1000	-	-	RNP1	RW19
20	DF	DUMIL	-	-	-	-	R	-	-	-	RNP1	RW19
30	TF	KAPOH	-	024 (020.8)	+3	20.0	-	-	-	-	RNP1	RW19
40	TF	SAKIS	-	007 (004.4)	+3	25.0	-	-	-	-	RNP1	RW19

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LAOAG/Laoag Intl (RPLI)
RWY 01/19
ALDIS1 ESSYL1 MELAN1
MINIO1 PAMKI1 SAKIS1



ARRIVAL ROUTE DESCRIPTION

STAR	ROUTE	PROCEDURE	REMARKS
ALDIS1	ALDIS - EMSEL - LAO DVOR/DME	From ALDIS on LAO DVOR/ DME Radial 300, to EMSEL at or above 7000 FT, to LAO DVOR/ DME at or above 4000 FT.	
ESSYL1	ESSYL - DULUD - PAYAS - LAO DVOR/DME	From ESSYL on LAO DVOR/ DME Radial 173, to DULUD at or above FL130, to PAYAS at or above 7000 FT, to LAO DVOR/ DME at or above 4000 FT.	
MELAN1	MELAN - KINLA - CURIM - LAO DVOR/DME	From MELAN on LAO DVOR/ DME Radial 189, to KINLA, to CURIM at or above 7000 FT, to LAO DVOR/DME at or above 4000 FT.	
MINIO1	MINIO - TADAO - LAO DVOR/DME	From MINIO on LAO DVOR/ DME Radial 037, to TADAO at or above 7000 FT, to LAO DVOR/ DME at or above 4000 FT.	
PAMKI1	PAMKI - LEMAP - LAO DVOR/DME	From PAMKI on LAO DVOR/ DME Radial 140, to LEMAP at or above 9000 FT, to LAO DVOR/ DME at or above 4000 FT.	
SAKIS1	SAKIS - KAPOH - LAO DVOR/DME	From SAKIS on LAO DVOR/ DME Radial 007, to KAPOH at or above 7000 FT, to LAO DVOR/ DME at or above 4000 FT.	

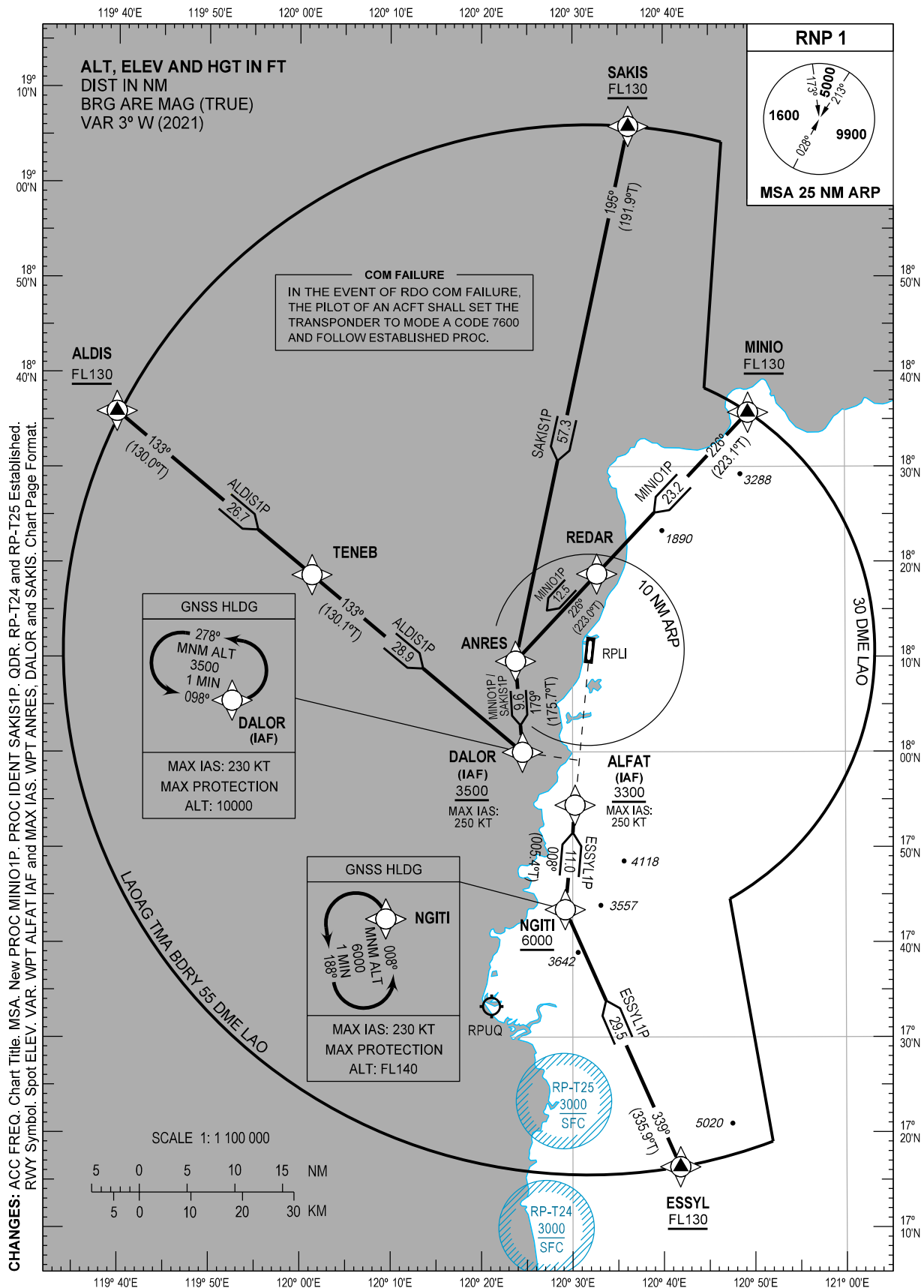
CHANGES: PROC and RTE Description. Page Format.

STANDARD ARRIVAL
CHART - INSTRUMENT
(STAR) - ICAO

TRANSITION
ALTITUDE
11000 FT

ACC - 119.3 / 128.7 (NORTHWEST)
- 120.5 / 119.3 (CENTRAL NORTH)
- 8942 / 5655 (SEA-2 HF)
APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)
RNP STAR RWY 01
ALDIS1P ESSYL1P
MINIO1P SAKIS1P



ARRIVAL ROUTE DESCRIPTION

STAR	ROUTE	PROCEDURE	REMARKS
ALDIS1P	ALDIS - TENEB - DALOR	From ALDIS at or above FL130, to TENEB, to DALOR at or above 3500 FT.	
ESSYL1P	ESSYL - NGITI - ALFAT	From ESSYL at or above FL130, to NGITI at or above 6000 FT, to ALFAT at or above 3300 FT.	
MINIO1P	MINIO - REDAR - ANRES - DALOR	From MINIO at or above FL130, to REDAR, to ANRES, to DALOR at or above 3500 FT.	
SAKIS1P	SAKIS - ANRES - DALOR	From SAKIS at or above FL130, to ANRES, to DALOR at or above 3500 FT.	

WAYPOINT LIST

Waypoint Name	Latitude	Longitude	Type
ALDIS	183552.4N	1193949.0E	Fly-by
ALFAT	175423.8N	1203015.9E	Fly-by
ANRES	180932.7N	1202344.0E	Fly-by
DALOR	175957.7N	1202429.5E	Fly-by
ESSYL	171621.3N	1204147.6E	Fly-by
MINIO	183543.9N	1204920.0E	Fly-by
NGITI	174324.1N	1202910.9E	Fly-by
REDAR	181842.6N	1203240.4E	Fly-by
SAKIS	190546.9N	1203609.9E	Fly-by
TENEB	181838.1N	1200119.7E	Fly-by

PROPOSED CODING FOR ARRIVAL RNP STAR RWY 01

Table 1: ALDIS1P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	ALDIS	-	-	-	-	-	+FL130	-	-	RNP1	RW01
20	TF	TENEB	-	133 (130.0)	+3	26.7	-	-	-	-	RNP1	RW01
30	TF	DALOR	-	133 (130.1)	+3	28.9	-	+3500	-250	-	RNP1	RW01

Table 2: ESSYL1P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	ESSYL	-	-	-	-	-	+FL130	-	-	RNP1	RW01
20	TF	NGITI	-	339 (335.9)	+3	29.5	-	+6000	-	-	RNP1	RW01
30	TF	ALFAT	-	008 (005.4)	+3	11.0	-	+3300	-250	-	RNP1	RW01

CHANGES: New PROC MINIO1P. RTE and PROC Description. WPT List. Proposed Coding. Page Format.

Table 3: MINIO1P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	MINIO	-	-	-	-	-	+FL130	-	-	RNP1	RW01
20	TF	REDAR	-	226 (223.1)	+3	23.2	-	-	-	-	RNP1	RW01
30	TF	ANRES	-	226 (223.0)	+3	12.5	-	-	-	-	RNP1	RW01
40	TF	DALOR	-	179 (175.7)	+3	9.6	-	+3500	-250	-	RNP1	RW01

Table 4: SAKIS1P

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	SAKIS	-	-	-	-	-	+FL130	-	-	RNP1	RW01
20	TF	ANRES	-	195 (191.9)	+3	57.3	-	-	-	-	RNP1	RW01
30	TF	DALOR	-	179 (175.7)	+3	9.6	-	+3500	-250	-	RNP1	RW01

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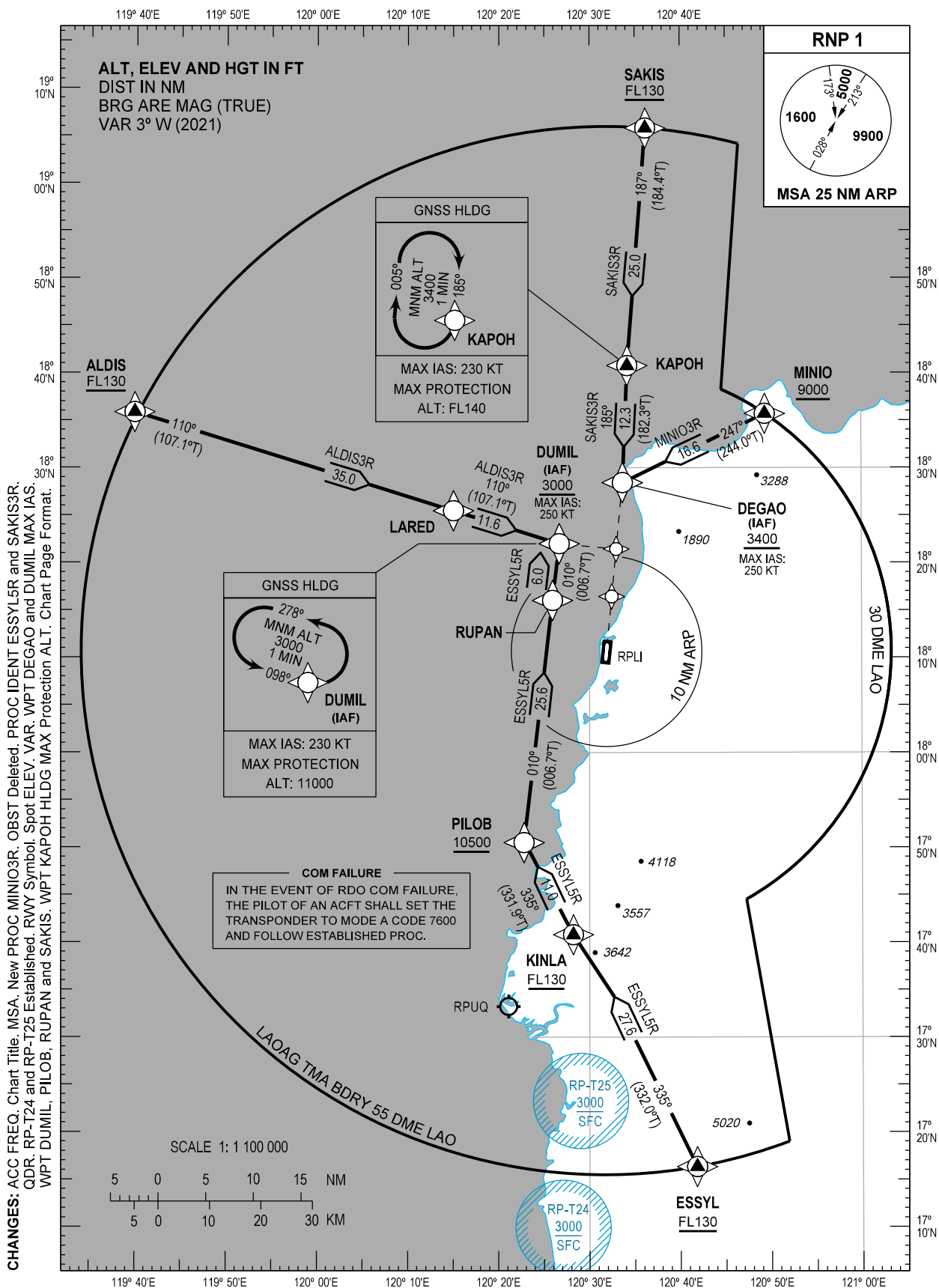
STANDARD ARRIVAL
CHART - INSTRUMENT
(STAR) - ICAO

TRANSITION
ALTITUDE
11000 FT

ACC - 119.3 / 128.7 (NORTHWEST)
- 120.5 / 119.3 (CENTRAL NORTH)
- 8942 / 5655 (SEA-2 HF)
APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)
RNP STAR RWY 19

ALDIS3R ESSYL5R
MINIO3R SAKIS3R



CHANGES: ACC FREQ. Chart Title. MSA. New PROC MINIO3R. OBST Deleted. PROC IDENT ESSYL5R and SAKIS3R.
QDR. RP-T24 and RP-T25 Established. RWY Symbol. Spot ELEV. VAR. WPT DEGAO and DUMIL. MAX IAS.
WPT DUMIL, PILOB, RUPAN and SAKIS. WPT KAPOH HLDG MAX Protection ALT. Chart Page Format.

ARRIVAL ROUTE DESCRIPTION

STAR	ROUTE	PROCEDURE	REMARKS
ALDIS3R	ALDIS - LARED - DUMIL	From ALDIS at or above FL130, to LARED, to DUMIL at or above 3000 FT.	
ESSYL5R	ESSYL - KINLA - PILOB - RUPAN - DUMIL	From ESSYL to KINLA at or above FL130, to PILOB at or above 10500 FT, to RUPAN, to DUMIL at or above 3000 FT.	
MINIO3R	MINIO - DEGAO	From MINIO at or above 9000 FT, to DEGAO at or above 3400 FT.	
SAKIS3R	SAKIS - KAPOH - DEGAO	From SAKIS at or above FL130, to KAPOH, to DEGAO at or above 3400 FT.	

WAYPOINT LIST

Waypoint Name	Latitude	Longitude	Type
ALDIS	183552.4N	1193949.0E	Fly-by
DEGAO	182825.8N	1203338.3E	Fly-by
DUMIL	182200.0N	1202639.8E	Fly-by
ESSYL	171621.3N	1204147.6E	Fly-by
KAPOH	184045.5N	1203409.0E	Fly-by
KINLA	174048.1N	1202811.9E	Fly-by
LARED	182528.6N	1201459.9E	Fly-by
MINIO	183543.9N	1204920.0E	Fly-by
PILOB	175030.5N	1202247.0E	Fly-by
RUPAN	181601.1N	1202555.4E	Fly-by
SAKIS	190546.9N	1203609.9E	Fly-by

PROPOSED CODING FOR ARRIVAL RNP STAR RWY 19

Table 1: **ALDIS3R**

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	ALDIS	-	-	-	-	-	+FL130	-	-	RNP1	RW19
20	TF	LARED	-	110 (107.1)	+3	35.0	-	-	-	-	RNP1	RW19
30	TF	DUMIL	-	110 (107.3)	+3	11.6	-	+3000	-250	-	RNP1	RW19

Table 2: ESSYL5R

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	ESSYL	-	-	-	-	-	+FL130	-	-	RNP1	RW19
20	TF	KINLA	-	335 (332.0)	+3	27.6	-	+FL130	-	-	RNP1	RW19
30	TF	PILOB	-	335 (331.9)	+3	11.0	-	+10500	-	-	RNP1	RW19
40	TF	RUPAN	-	010 (006.7)	+3	25.6	-	-	-	-	RNP1	RW19
50	TF	DUMIL	-	010 (006.7)	+3	6.0	-	+3000	-250	-	RNP1	RW19

Table 3: MINIO3R

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	MINIO	-	-	-	-	-	+9000	-	-	RNP1	RW19
20	TF	DEGAO	-	247 (244.0)	+3	16.6	-	+3400	-250	-	RNP1	RW19

Table 4: SAKIS3R

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	SAKIS	-	-	-	-	-	+FL130	-	-	RNP1	RW19
20	TF	KAPOH	-	187 (184.4)	+3	25.0	-	-	-	-	RNP1	RW19
30	TF	DEGAO	-	185 (182.3)	+3	12.3	-	+3400	-250	-	RNP1	RW19

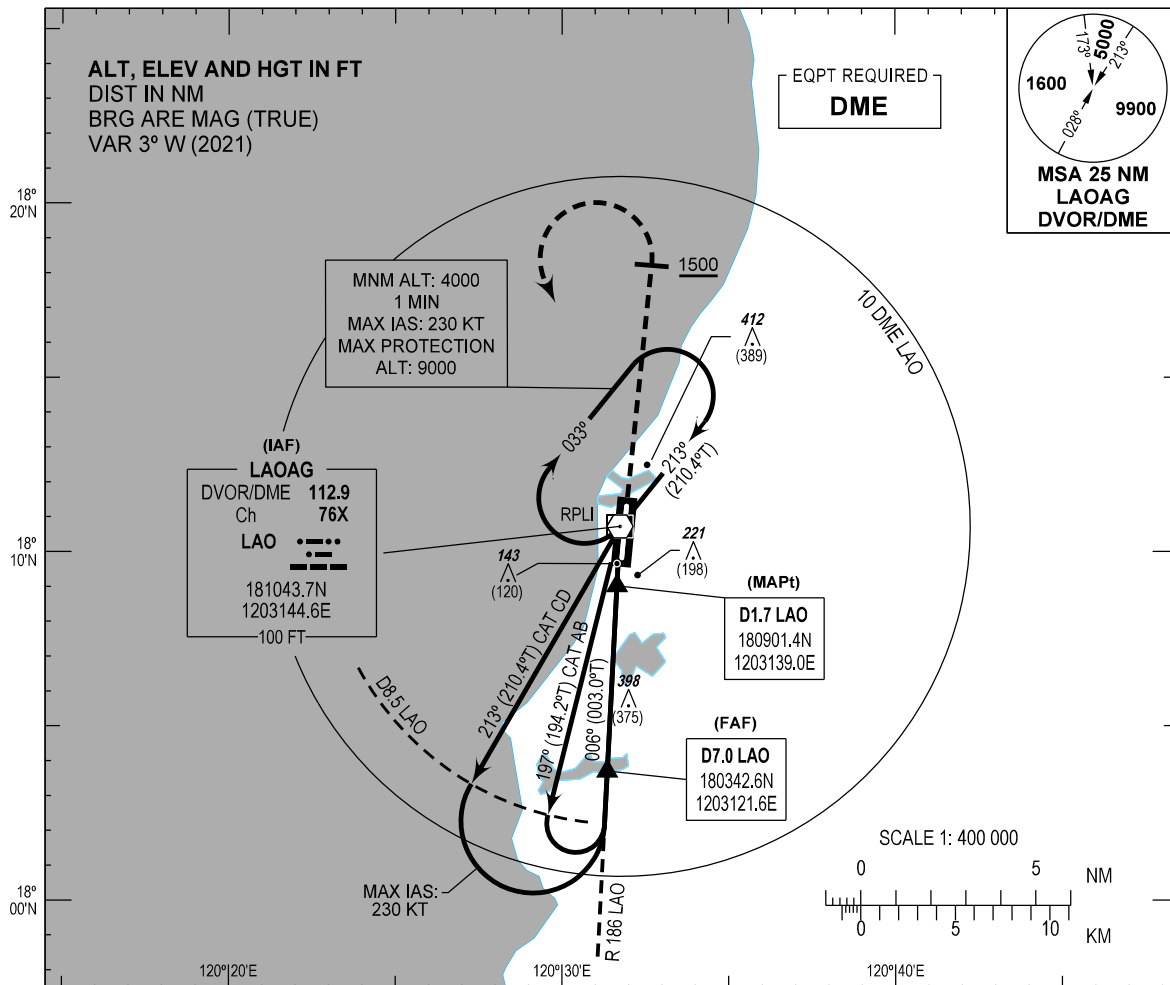
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INSTRUMENT
APPROACH
CHART - ICAO

AERODROME ELEV 24 FT
HEIGHTS RELATED TO
THR RWY 01 - ELEV 23 FT

APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)
VOR RWY 01



TRANSITION ALT : 11000
TRANSITION LVL : FL130

D7.0 LAO
(FAF)
2000

LAO DVOR/DME
(IAF)
4000

MISSED APPROACH

Straight-out climb to **1500 FT**, turn left direct to **LAO DVOR/DME** climbing to **4000 FT** and hold or as instructed by ATC. Do not turn before **LAO DVOR/DME**.

D8.5 LAO

D1.7 LAO

SDF

D3.5 LAO

900

*MAX IAS for CAT D 200 KT

650 (627)

OCA (H)

THR ELEV 23
(THR RWY 01)

THR (NM)
LAO (NM)

6.1
7

3.5
3.5

2.6
1.7

0.8
0

0
0

OCA (H)	A	B	C	D
STRAIGHT-IN	480 (457)			
CIRCLING*	710 (686)		810 (786)	

NEW CHART

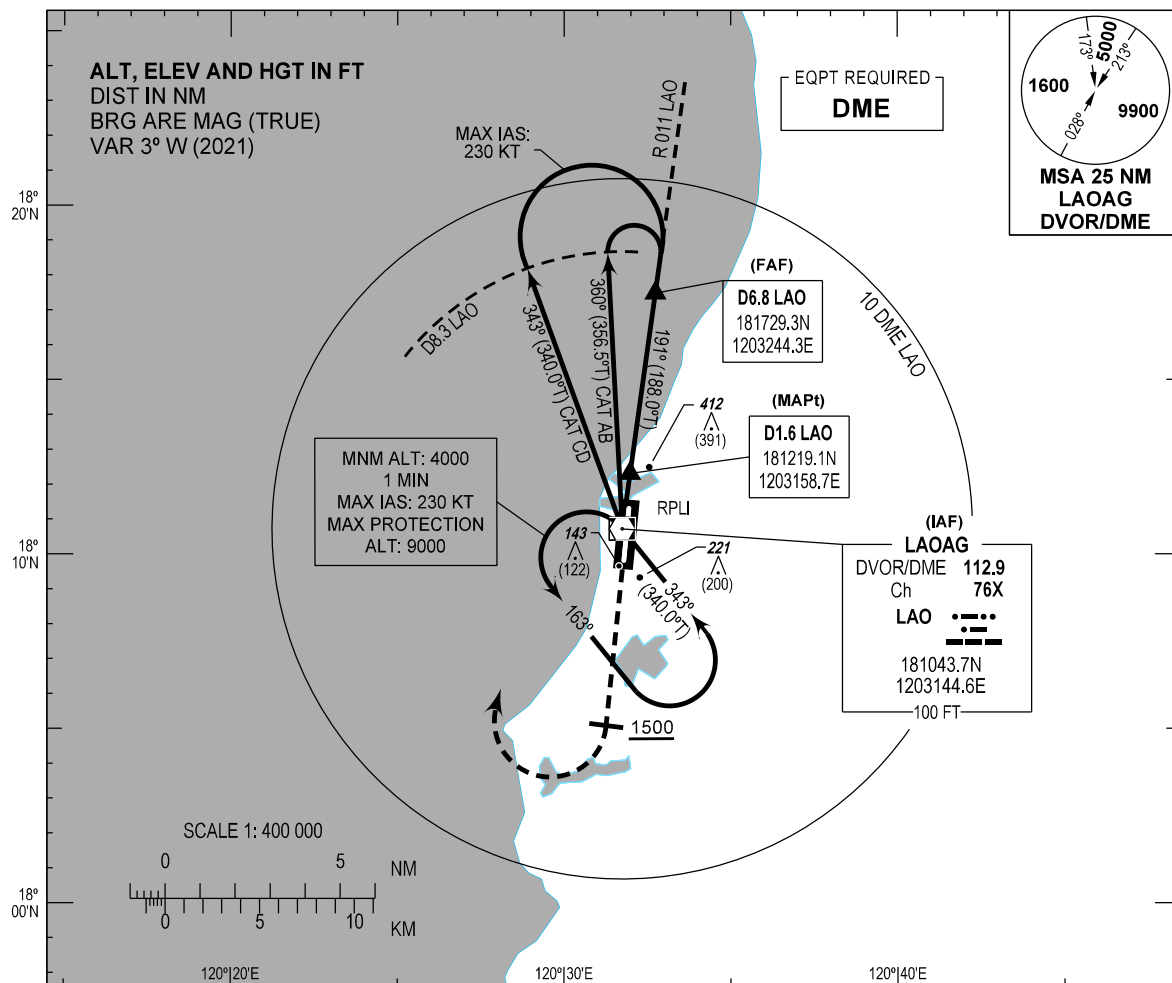
NM TO LAO	7	6	5	4	3	2.2
ALT (HGT)	2000 (1977)	1690 (1667)	1370 (1347)	1060 (1037)	740 (717)	480 (457)

INSTRUMENT
APPROACH
CHART - ICAO

AERODROME ELEV 24 FT
HEIGHTS RELATED TO
THR RWY 19 - ELEV 21 FT

APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)
VOR RWY 19



MISSED APPROACH

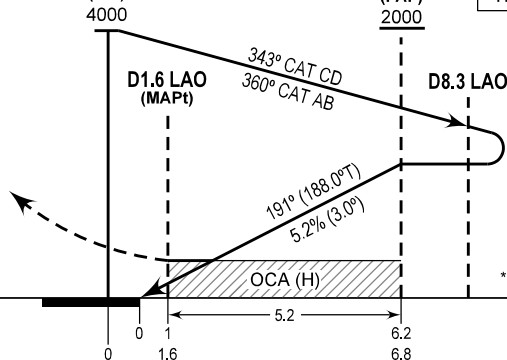
Straight-out climb to **1500 FT**, turn right direct to **LAO DVOR/DME** climbing to **4000 FT** and hold or as instructed by ATC. Do not turn before **LAO DVOR/DME**.

LAO DVOR/DME

D6.8 LAO

TRANSITION ALT : 11000
TRANSITION LVL : FL130

THR ELEV 21
(THR RWY 19)



OCA (H)	A	B	C	D
STRAIGHT-IN	660 (639)			
CIRCLING*	710 (686)		810 (786)	

NM TO LAO	7	6	5	4	3	2.5
ALT (HGT)	2000 (1979)	1750 (1729)	1440 (1419)	1130 (1109)	820 (799)	660 (639)

INSTRUMENT
APPROACH
CHART - ICAO

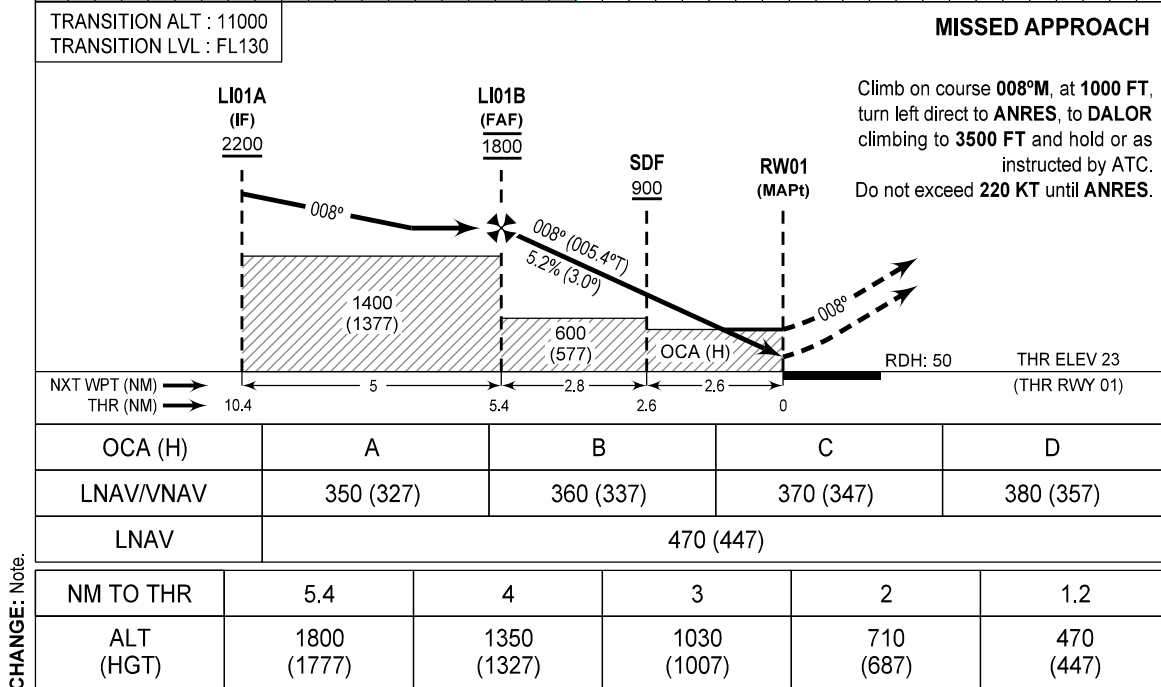
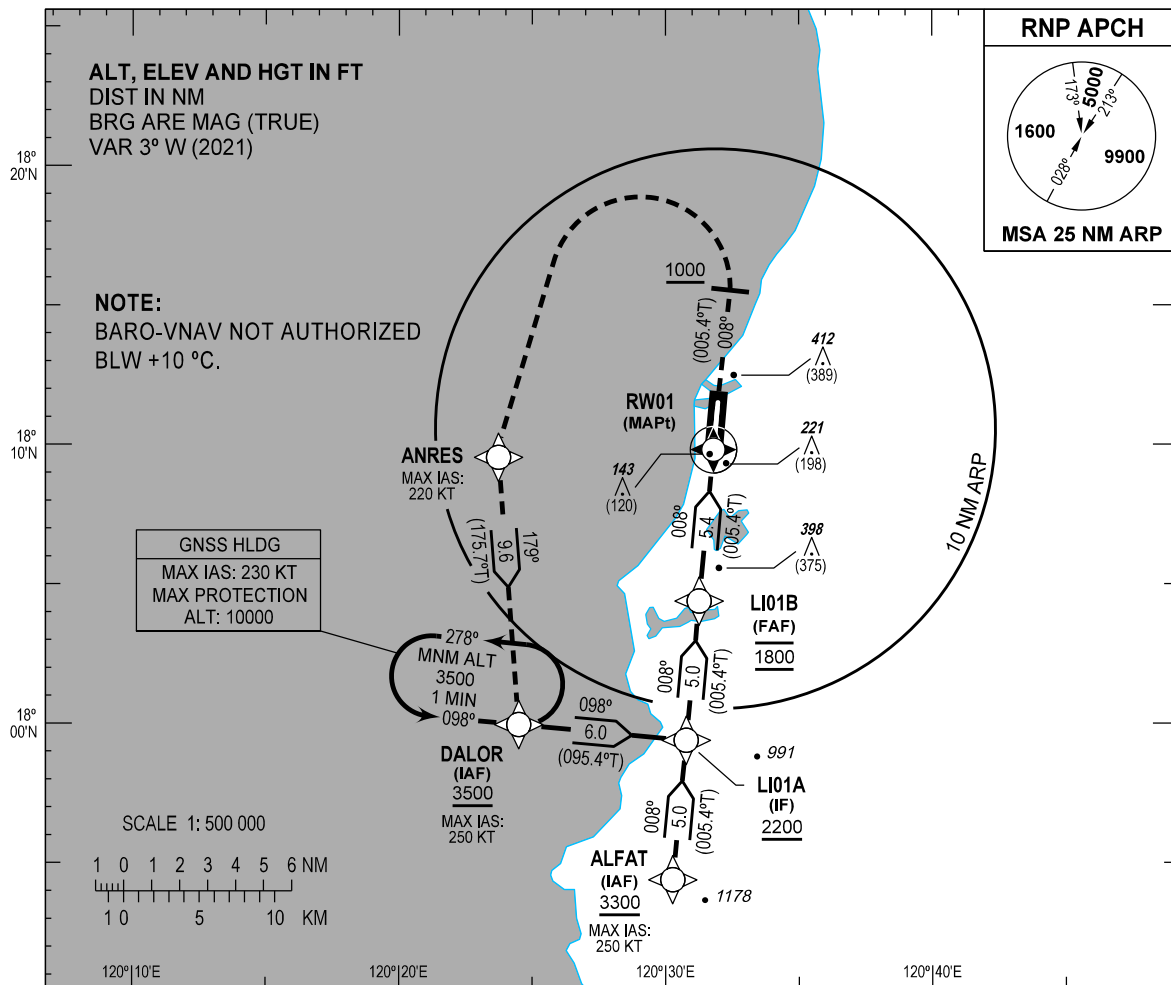
AERODROME ELEV 24 FT

HEIGHTS RELATED TO
THR RWY 01 - ELEV 23 FT

APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)

RNP RWY 01



WAYPOINT LIST

Waypoint Name	Latitude	Longitude	Type
ALFAT	175423.8N	1203015.9E	Fly-by
ANRES	180932.7N	1202344.0E	Fly-by
DALOR	175957.7N	1202429.5E	Fly-by
LI01A	175923.7N	1203045.6E	Fly-by
LI01B	180423.5N	1203115.3E	Fly-by
RW01	180948.99N	1203147.47E	Flyover

PROPOSED CODING FOR INSTRUMENT APPROACH RNP RWY 01

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	ALFAT	-	-	-	-	-	+3300	-250	-	RNP APCH	ALFAT
20	TF	LI01A	-	008 (005.4)	+3	5.0	-	+2200	-	-	RNP APCH	ALFAT
10	IF	DALOR	-	-	-	-	-	+3500	-250	-	RNP APCH	DALOR
20	TF	LI01A	-	098 (095.4)	+3	6.0	-	+2200	-	-	RNP APCH	DALOR
10	IF	LI01A	-	-	-	-	-	+2200	-	-	RNP APCH	-
20	TF	LI01B	-	008 (005.4)	+3	5.0	-	@1800	-	-	RNP APCH	-
30	TF	RW01	Y	008 (005.4)	+3	5.4	-	@73	-	-3.0/-	RNP APCH	-
10	CA	-	-	008 (005.4)	+3	-	-	+1000	-	-	RNP APCH	-
20	DF	ANRES	-	-	-	-	L	-	-220	-	RNP APCH	-
30	TF	DALOR	-	179 (175.7)	+3	9.6	-	+3500	-250	-	RNP APCH	-

INSTRUMENT
APPROACH
CHART - ICAO

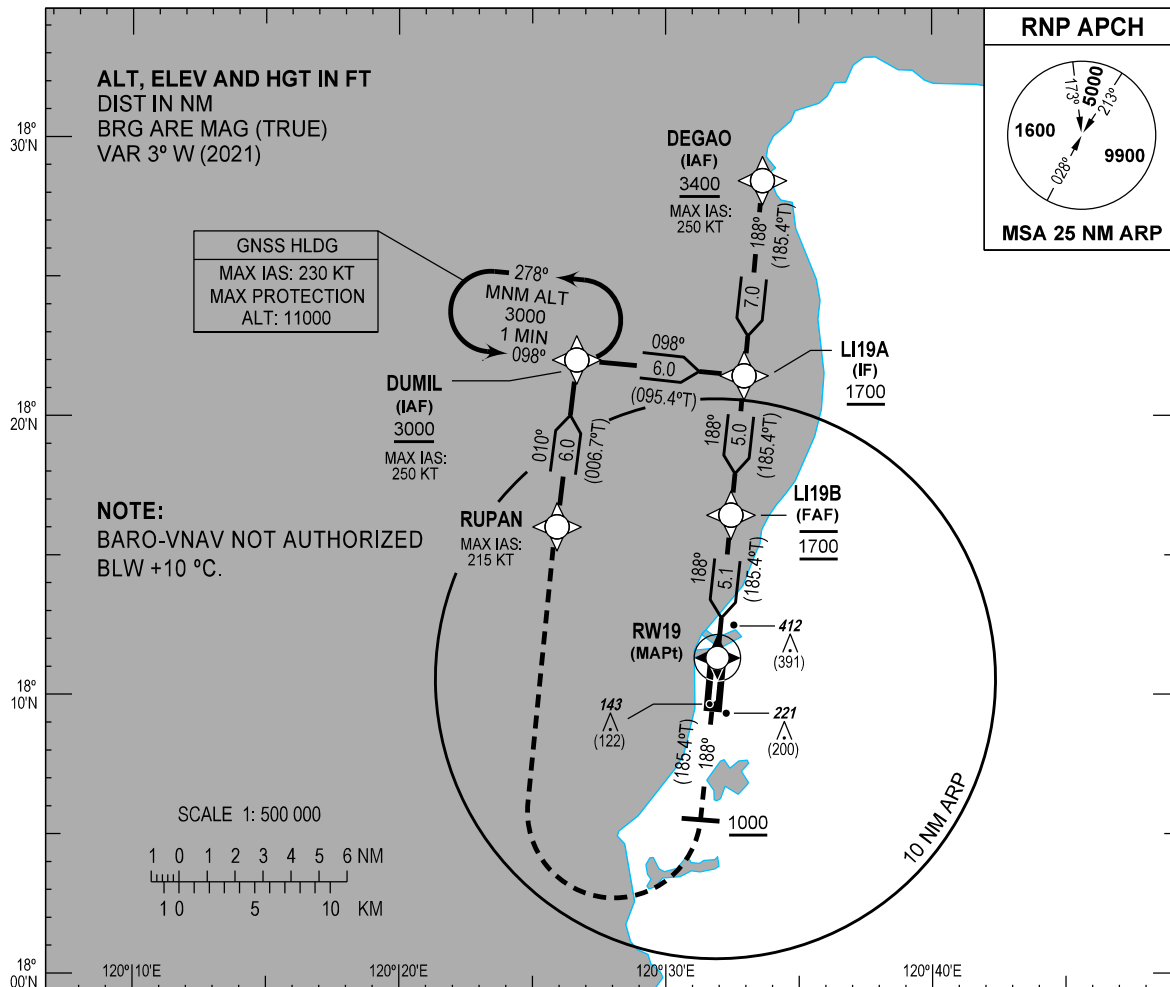
AERODROME ELEV 24 FT

HEIGHTS RELATED TO
THR RWY 19 - ELEV 21 FT

APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)

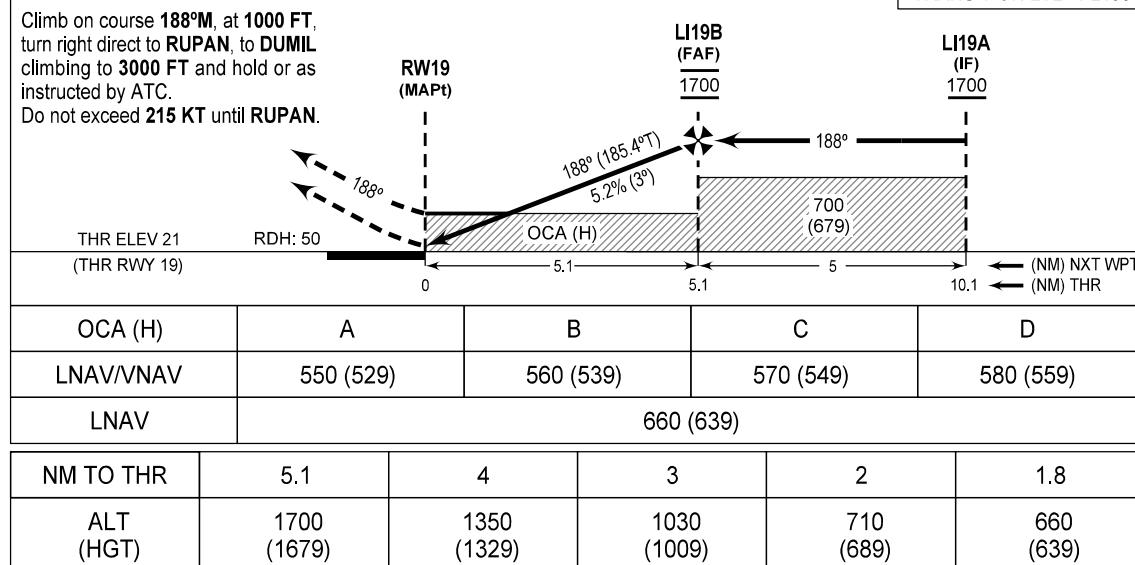
RNP RWY 19



MISSED APPROACH

Climb on course 188°M, at 1000 FT,
turn right direct to RUPAN, to DUMIL
climbing to 3000 FT and hold or as
instructed by ATC.
Do not exceed 215 KT until RUPAN.

TRANSITION ALT : 11000
TRANSITION LVL : FL130



WAYPOINT LIST

Waypoint Name	Latitude	Longitude	Type
DEGAO	182825.8N	1203338.3E	Fly-by
DUMIL	182200.0N	1202639.8E	Fly-by
LI19A	182126.0N	1203256.6E	Fly-by
LI19B	181626.2N	1203226.8E	Fly-by
RUPAN	181601.1N	1202555.4E	Fly-by
RW19	181119.14N	1203156.40E	Flyover

PROPOSED CODING FOR INSTRUMENT APPROACH RNP RWY 19

Serial Number	Path Descriptor	Waypoint Identifier	Flyover	Course °M (°T)	Magnetic Variation (°)	Distance (NM)	Turn Direction	Altitude (FT)	Speed limit (KT)	VPA/TCH	Navigation Specification	Transition Identifier
10	IF	DEGAO	-	-	-	-	-	+3400	-250	-	RNP APCH	DEGAO
20	TF	LI19A	-	188 (185.4)	+3	7.0	-	+1700	-	-	RNP APCH	DEGAO
10	IF	DUMIL	-	-	-	-	-	+3000	-250	-	RNP APCH	DUMIL
20	TF	LI19A	-	098 (095.4)	+3	6.0	-	+1700	-	-	RNP APCH	DUMIL
10	IF	LI19A	-	-	-	-	-	+1700	-	-	RNP APCH	-
20	TF	LI19B	-	188 (185.4)	+3	5.0	-	@1700	-	-	RNP APCH	-
30	TF	RW19	Y	188 (185.4)	+3	5.1	-	@71	-	-3.0/-	RNP APCH	-
10	CA	-	-	188 (185.4)	+3	-	-	+1000	-	-	RNP APCH	-
20	DF	RUPAN	-	-	-	-	R	-	-215	-	RNP APCH	-
30	TF	DUMIL	-	010 (006.7)	+3	6.0	-	+3000	-250	-	RNP APCH	-

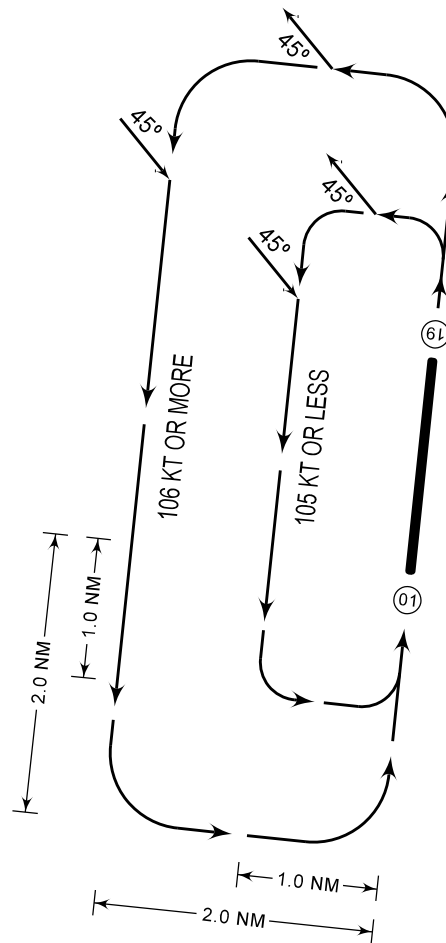
TRAFFIC
CIRCUIT
CHART

AERODROME ELEV
24 FT

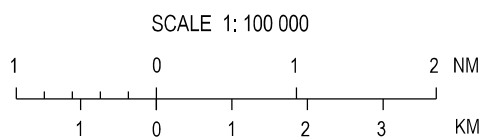
APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)
RWY 01

CHANGES: ACFT, ALT, VAR and Units of Measurement Deleted. Data Resolution. Chart Page Format.



LEFT



PROCEDURE

- ARRIVING AIRCRAFT SHALL ENTER THE TRAFFIC CIRCUIT ON THE DOWNWIND LEG AT AN ANGLE OF 45 DEGREES.
- DEPARTING AIRCRAFT SHALL FOLLOW THE TRAFFIC CIRCUIT AND LEAVE THE CIRCUIT AT AN ANGLE OF 45 DEGREES.

TRAFFIC
CIRCUIT
CHART

AERODROME ELEV
24 FT

APP - 122.3
TWR - 118.1

LAOAG/Laoag Intl (RPLI)
RWY 19

CHANGES: ACFT, ALT, VAR and Units of Measurement Deleted. Data Resolution. Chart Page Format.

